Imperial College London

Pathways to Opioid Relief: Behavioural Interventions to Adress Opioid Misuse Among U.S. Veterans in Rural Central-Appalachia

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Background and Problem Identification

1.1 The Opioid Epidemic

The opioid epidemic remains a pressing public health crisis in the United States, with a mortality burden among the most severe in the world (1,2). In the 1990s, aggressive marketing of prescription opioids such as oxycodone and hydrocodone as a means for physicians to adequately treat chronic pain spurred the first wave of the epidemic, leading to widespread misuse and addiction (1,3). By 2010, prescription rates had surged to 81.2 prescriptions per 100 persons, enough to medicate every adult American throughout an entire month (3). As public scrutiny and regulatory oversight increased, a user shift towards the cheaper and more readily available heroin marked the second wave, culminating in a surge of heroin-related deaths peaking in 2016 (3). This was succeeded by a third wave, characterized by synthetic opioids like fentanyl, whose potency led to a 1,059% increase in synthetic opioid-related deaths between 2013 and 2019 (3). By 2019, synthetic opioids were responsible for 73% of all opioid-related deaths (3).

While national opioid prescription rates have since declined by 53.8% to 37.5 per 100 in 2023, the epidemic's impact in terms of overdose deaths remains severe (4–6). In 2021, 1 in 22 deaths in the U.S. was attributable to unintentional opioid-toxicity, underscoring the persistent societal burden of these substances (7).

Constituting ~6% of the country's adult population, the crisis has disproportionately affected military veterans, who face psychological, physiological, and socio-structural vulnerabilities stemming from barriers to accessing care (8–11). Notably, a systematic-review by the U.S. Department of Veterans Affairs (VA) found that veterans are twice as likely to die from opioid-overdose as the general population, further highlighting their susceptibility to the adverse effects opioid-prescription and misuse (12).

1.2 Target Population

While the opioid epidemic has affected the U.S. broadly, the South, particularly the Appalachian region, has been identified by the Centres for Disease Control (CDC) as a critical epicentre of the opioid epidemic (Appendix A) (4,13–15). Appalachia, spanning parts of 13 states, faces elevated prescription and overdose death rates relative to the national average (Appendix B)(4,13,16). In 2023, Kentucky's prescription rate was 68.2 per 100 residents, and West-Virginia had the highest overdose death rate in the nation (4,17). Such member states of Central-Appalachia are particularly affected, with rural areas experiencing an inordinate burden owing to socio-economic vulnerabilities and limited healthcare infrastructure (18,19). Veterans make up ~17.4% of these areas, facing unique challenges including limited access to alternative therapies, long distances to VA

facilities, and socio-economic factors that contribute to systemic poverty and substance abuse (19–21).

Compounding these issues, many veterans rely heavily on opioids to manage chronic pain stemming from service-related injuries (9–11). Meta-analysis by Qureshi et al. indicated that a prevalence of up to ~51% of veterans in VA care report chronic pain severe enough to require medical treatment, for which prescription-opioids have often been the first-line therapy (9,22). While effective against pain, prolonged use carries notable risks of dependence, misuse, and eventual overdose (9). These risks are exacerbated by the co-occurrence of PTSD, which affects up to a lifetime prevalence of 7% in all veterans, and contributes to both psychological and pain chronicity, creating a cycle of dependency that is difficult to break (23). Roughly 25% of veterans prescribed opioids are speculated to misuse them, ~10% of which develop an opioid-use disorder (OUD) (24). These vulnerabilities underscore the urgency of addressing opioid-toxicity among veterans.

1.3 Existing Interventions

The Veterans Health Administration (VHA) has addressed this nation-wide issue through implementation of several initiatives to combat both prescription and illicit opioid-misuse among veterans (25,26). The Opioid Safety initiative (OSI), introduced in 2013, has achieved a 64% reduction in opioid-prescription across the VA healthcare system through stricter guidelines and promotion of alternative therapies such as cognitive behavioural therapy (CBT), physical therapy, and medications like methadone and buprenorphine (25,26).

Despite these achievements, the OSI faces substantial challenges in rural America, including in rural Central-Appalachia (19,21). Veterans in these areas often struggle to access non-opioid treatments promoted by the initiative due to geographical barriers, therapy provider shortages, and limited telehealth infrastructure (19,21). Additionally, stigma surrounding OUDs and mental health further deter many veterans from seeking help (23,27). Systemic issues, such as long waiting times at VA facilities and fragmented care co-ordination, only compound these challenges (27).

At the state level, Prescription Drug Monitoring Programs (PDMPs) have demonstrated some success in reducing opioid-misuse, particularly in Kentucky and West-Virginia (28,29). However, their effectiveness is limited in addressing the underlying behavioural and structural factors driving opioid dependency among veterans (30,31).

1.4 Problem Statement and Target Behaviour

The intersection of behavioural, structural, and systemic factors therefore makes opioid misuse among veterans in rural Central-Appalachia a complex public health issue. These limitations accentuate the need for interventions that address behavioural drivers among veterans contributing to misuse behaviours and promote evidence-based non-opioid pain-management strategies.

The target-behaviour for this behavioural analysis is therefore **encouraging U.S. veterans in the rural Central-Appalachian region to adopt non-opioid pain-management strategies**, thereby reducing reliance on prescription and illicit opioids.

Behavioural Analysis

2.1 Conceptualizing Behavioural Predictors

Theoretical frameworks like the COM-B model can be applied to identify predictors to behaviour (Appendix C) (32). Specifically, COM-B conceptualizes ways in which Capability, Opportunity, and Motivation interact to influence behaviour (32). Through acknowledgement of barriers to the adoption of non-opioid management in rural Central-Appalachian veterans, facilitators that may contribute to uptake of the target-behaviour can be elucidated and used to guide intervention development.

Capability, **physical** (***C-Ph***) and **psychological** (***P-Ps***), have been known to influence health-behaviours in veterans (24). A national census survey by Mulcahy *et al.* found that a staggering ~52% of veterans reported some form of physical disability between 2015-2018 (33). While not a direct limitation for most, presence of co-occurring physical impairment could depreciate **reflexive motivational** (***M-Re***) beliefs about self-efficacy regarding accessing care. In rural areas of the Central-Appalachians, veterans often express a lack of awareness about non-opioid strategies offered by the OSI or PDMPs, consistent with qualitative insights in other areas of rural America (19,34). Moreover, low levels of digital literacy and misconceptions about efficacies of these alternatives in veterans who *are* aware have been found to reinforce a preference of opioids, consciously motivating them toward misuse behaviours (34).

Physical (***O-Ph***) and **social opportunity** (***O-So***) have also been found to affect veteran behaviours in the Appalachians (35). A reoccurring theme emerging across multiple qualitative research efforts is that limited access to mental-health, addiction, and specialized care centres (such as CBT, physical therapy, telehealth, and integrative health practices), coupled with a shortage of trained providers and poor transportation services contribute greatly to the challenges faced by veterans in rural Central-Appalachia motivated to seek help (19,35,36). While free coverage of VHA and VA rehabilitation and care mean financial challenges are not a primary concern among

veterans, socio-economic deprivation in rural Appalachian communities has created social environments where veterans are likely to turn to substance abuse as a coping mechanism against hopelessness and stress (19,35,37). Moreover, the social stigmas around mental care and OUDs rooted in residual beliefs from the military culture of self-reliance and stoicism further discourage open discussion with care providers. **Automatic motivational** (***M-Au***) feelings of fear and frustration surrounding pain and withdrawal have been cited across studies, with veterans often describing opioids as both a 'lifeline' and a 'trap' (38).

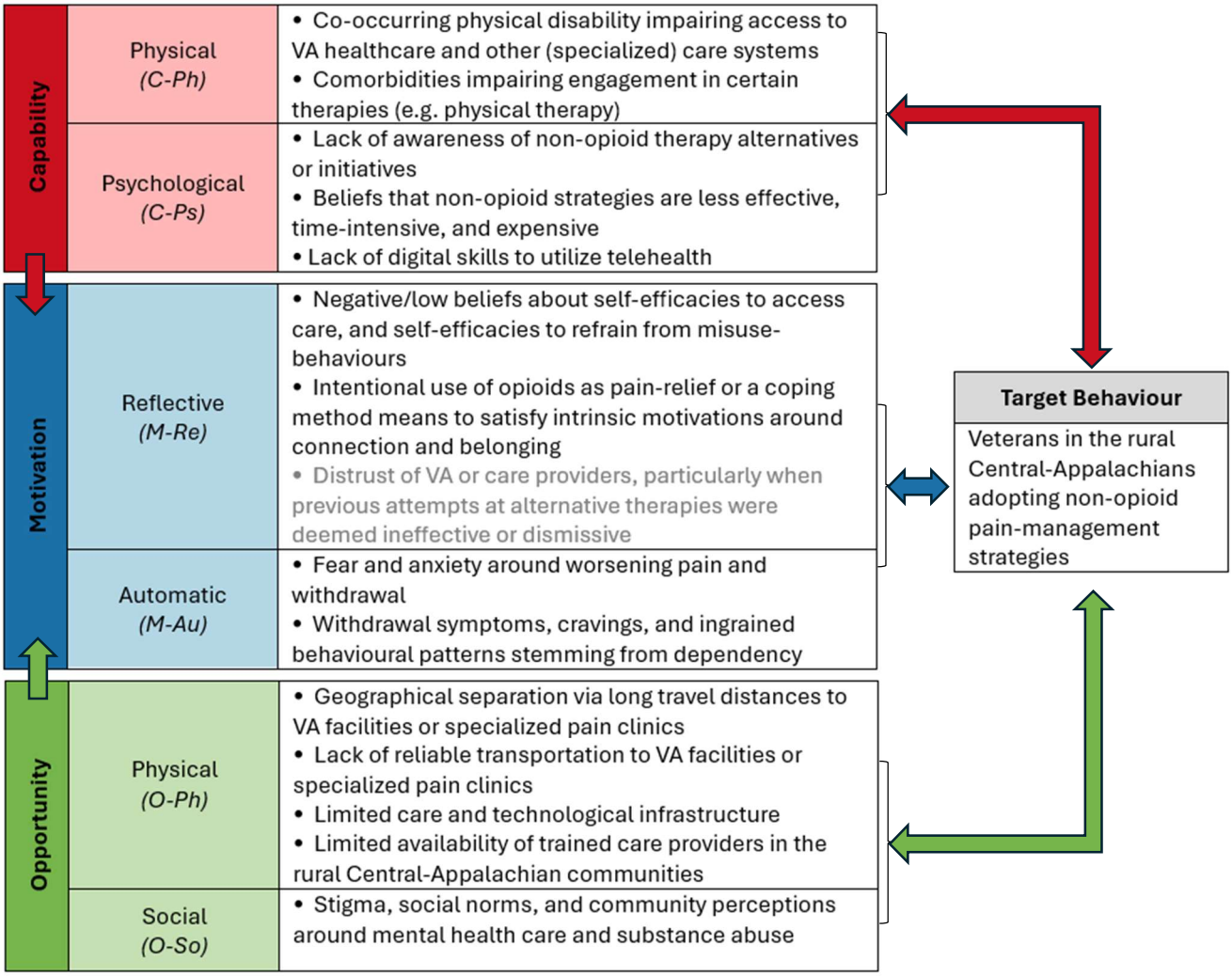


Figure 1: *Identified barriers to adoption of non-opioid pain-management strategies among veterans in the rural Central-Appalachians by COM-B application.*

The stubborn beliefs and trust in opioid-use among veterans could owe in part to broader trends in the medical community advocating opioid use as an analgesic with 'low abuse potential' preceding the first wave of the epidemic (3). Through inadvertent use of nudge-based principles, habitual use of prescription-opioids empowered by messenger, norm, and default MINDSPACE considerations may still drive the establishment of physiological dependence and addiction. Rapid pain-relief and euphoric effects offered by opioids also

pose serious motivational barriers to the adoption of non-opioid management through incentivisation and affect. Summatively, COM-B considerations suggest that barriers exist on individual, interpersonal, organizational, communal, and systemic levels.

The Theoretical Domains Framework (TDF) allows for a more granular understanding of behavioural determinants by identifying what factors that can change require positive change to facilitate uptake of the target-behaviour in the target-population (Appendix D) (39). Application of the TDF has identified **skills**, **knowledge**, **behavioural regulation**, **intentions**, **optimism**, **beliefs about capabilities** and **consequences**, **emotion**, **environmental context**, and **social influences** as domains which can be targeted to implement behavioural change.

COM-B Subdomain	TDF Domain	Relevance – Addressing Behavioural Barriers
C-Ph	Skills	Facilitate improved digital literacy and technical skills
C-Ps	Knowledge	Boost awareness of the existence and efficacies of non-opioid pain management strategies
		Facilitate improved health literacy around opioid use
	Memory, Attention, and Decision Processes	None identified
	Behavioural Regulation	Address dependence and/or habitual use of prescription and illicit opioids
M-Re	Intentions	Address intention to avoid non-opioid alternatives (in some) due to misconceptions and stigmas
	Goals	None identified
	Optimism	Address lack of optimism due to reflective processes triggered by feelings of fear, frustration, and hopelessness
	Beliefs about Capabilities	Encourage improvement of self-efficacy about ability to access VA or other care facilities
		Encourage improvement of self-efficacy about ability to abstain from opioid-use in addicted veterans
	Beliefs about Consequences	Address beliefs that non-opioid therapies are insufficient in effectiveness
	Social / Professional Role and Identity	None identified
M-Au	Reinforcement	None identified
	Emotion	Address vulnerability to implicit incentivisation and affect provided by opioid use
		Address anxiety about withdrawal symptoms and fear of inadequate pain relief without opioids
		Address feelings of frustration, hopelessness, and trappage from dependence and addiction
O-Ph	Environmental Context and Resources	Boost ability to access healthcare (due to inadequate healthcare infrastructure and transportation)
O-So	Social Influences	Reduce/limit susceptibility to negative influence of stigmas and societal influences

Figure 2: *Application of the Theoretical Domains Framework to identify factors that require positive change to facilitate uptake of the target-behaviour in the target-population.*

2.2 Evaluating Intervention Functions

Using the Behavioural Change Wheel (BCW), relevant COM-B/TDF domains can then be matched systematically to nine possible intervention functions (Appendix E) (32).

COM-B Subdomain	TDF Domain	BCW Intervention Functions								
		Education	Persuasion	Incentivisation	Coercion	Training	Restriction	Environmental Restructuring	Modelling	Enablement
C-Ph	Skills	x				x				
C-Ps	Knowledge	x								
	Behavioural Regulation					x				x
M-Re	Intentions	x	x	x	x					
	Optimism	x	x							
	Beliefs about capabilities	x	x	x						
	Beliefs about consequences	x	x							
M-Au	Emotion		x	x	x				x	x
O-Ph	Environmental context							x		x
O-So	Social influences						x	x		x

Figure 3: *Links between COM-B, TDF, and possible BCW intervention functions. Coloured blocks indicate links between COM-B and intervention functions. Blocks marked 'x' represent contextually relevant links between TDF and intervention functions identified in analysis.*

Figure 3 demonstrates how **education**, **persuasion**, and **enablement** can contribute the most positive change toward the target-behaviour. Others, namely incentivisation, coercion, training, and restructuring warrant assessment through APEASE criteria, which considers acceptability, practicability, effectiveness, affordability, side-effects, and equity aspects of policy implementation (40). While restriction will be excluded due to low predicted impact, addressing key emotional motivational drivers underpinning misuse-behaviours in veterans mean **modelling** could facilitate significant positive change.

Intervention Function	Assessment of APEASE criteria in the context of encouraging U.S. veterans in the rural Central-Appalachians to adopt non-opioid pain-management strategies
Incentivisation	Can be perceived as patronizing to veterans, undermining trust and autonomy. May not be effective in sustaining long-term behaviour change and does not address underlying behavioural predictors. Both non-monetary and monetary incentives are costly.
Coercion	Neither acceptable to stakeholders, practicable, nor effective in the context of the positive target health behaviour. Coercion could facilitate psychological resistance, worsen trust, and increase disparity between advantaged and disadvantaged stares and counties of the Central-Appalachian region, characterized by a high percentage of rural populations (41).
Training	Essential when addressing practical skill gaps like that of digital literacy. Addresses an identified barrier to the target-behaviour in the target-population despite cost concerns.
Environmental Restructuring	Considerably costly in terms of material and/or human resources compared against other intervention functions. Wasteful; it does not take advantage of existing underutilized care structures.

Figure 4: *Assessment of BCW intervention functions using APEASE criteria.*

Due to these APEASE considerations, **training** will be included.

2.3 Exploring Policy Categories

The BCW can also be used to match each of the five identified intervention functions to seven possible policy categories, which can also be assessed using APEASE.

Policy Categories	Intervention Functions					Assessment of Policy Function compliance to APEASE criteria in the context of encouraging U.S. veterans in the rural Central-Appalachians to adopt non-opioid pain-management strategies
	Education	Persuasion	Training	Modelling	Enablement	
Communication / Marketing						High predicted relevance and effectiveness.
Guidelines						May be effective for care-providers but not veterans. Concerns around practicability (resources).
Fiscal						Low predicted relevance.
Regulation						Predicted to be ineffective regarding illicit opioid-use. Concerns around acceptability and practicability.
Legislation						Predicted to be ineffective regarding illicit opioid-use. Concerns around acceptability and practicability.
Environmental / Social planning						Low predicted relevance and effectiveness of environmental but not social planning.
Service provision						Highest predicted relevance and effectiveness. Some concerns around affordability and practicability (resources).

Figure 5: *Links between identified BCW intervention functions and possible policy categories.*

The analysis above suggests **communication/marketing**, **service provision**, and **environmental/social planning** are best suited to utilizing the identified intervention functions to facilitate positive behavioural change in the target-population. Use of these policy categories contributed greatly to the successful implementation of PDMPs in states like Kentucky and West-Virginia, ultimately contributing to improved health outcomes in their veteran populations (42,43). Collaborative efforts and targeted educational initiatives significantly increased the effectiveness of the programs in addressing the opioid epidemic in two of the most concerning member states of the Central-Appalachians, and America as a whole (4,17).

Proposing Interventions

Resting on the analysis and supported by the success of these states, the following interventions aim to encourage uptake of the target-behaviour in the target-population.

3.1 Digital Media Platform

The first intervention seeks to provide an easy-to-use, attractive, and social digital platform to educate veterans on non-opioid pain management while enhancing their digital skills and fostering a forum community. Accessible via phones, laptops, and other devices, the platform ensure veterans can use it anytime and anywhere, bypassing mobility and transportation concerns. Content will utilize gamification, quizzes, and interactive modules to make learning engaging and effective, while tailored educational material will address individual veterans' needs and provide feedback to reinforce learning. It will break down complex concepts into palatable, reminder-driven modules and tutorials to address poor health and digital literacy among veterans. This will include teaching veterans how to navigate services, video conferencing tools, and troubleshooting technical issues. Additionally, the platform will teach skills required to access existing telehealth services and inform veterans about care centres nearest to their location.

3.2 Peer Support Groups

Simultaneous implementation of peer support groups will foster supportive communities where veterans in rural Central-Appalachian communities can share experiences, strategies, and encouragement to manage chronic pain without opioids. These groups will offer in-person and virtual meetings led by rehabilitated veterans and supervised by trained facilitators who understand the unique challenges of veteran-specific pain-management. Non-judgemental and open discussion of personal stories and testimonials could present salience and relatability, addressing issues of self-efficacies, beliefs capabilities, and social stigmas by providing a new outlet for veterans' sense of

camaraderie. To encourage consistent participation, non-monetary incentives including recognition certificates and badges will be awarded for consistent attendance and contribution. Incentivisation, in this context, will provide opportunities for success that can lead to mastery experiences, the most powerful source of self-efficacy (44).

Intervention	**Digital Media Platform**	**Peer Support Groups**
Description	Online media hub with educational, locational, and scheduling reminder-based functions	Support groups where peers share experiences transitioning to non-opioid therapies and learn from and with each other
Aim	Address locational awareness and digital and health literacy	Reduce stigma and promote social learning through trusted and relatable community voices
Employed Behavioural Nudges	Ease of access; attractiveness; sociality	Role modelling and social proof; incentivisation; affect through validation; ease of access; commitment
Policy Category	Communication/Marketing Service Provision	Communication/Marketing Service Provision Environmental/Social Planning
Intervention Function	Education; Training; Enablement	Education; Persuasion; Modelling; Enablement
TDF Change Component	Skills; knowledge	Knowledge; Behavioural regulation; Intentions; Optimism; Beliefs about capabilities and consequences; Emotion; Social influences
COM-B Subdomain	*C-Ph* and *C-Ps*	*C-Ps*; *M-Re*; *M-Au*; *O-So*

Figure 6: *Proposed interventions to encourage uptake of non-opioid pain-management therapies among veterans in rural Central-Appalachia. Successful implementation of both would address O-Ph concerns.*

Successful implementation of these interventions could significantly enhance veteran engagement with non-opioid therapies while addressing key underlying predictors of opioid-misuse. Analysis findings are consistent with existing research which highlights the effectiveness of peer-led groups in reducing substance abuse and relapse rates, improving treatment-retention, and strengthening relationships with providers and social supporters (45–47). Addressing digital literacy is critical, as it facilitates access to telehealth, offering personalized, consistent, remote, and comparatively cost-effective care that surpasses the reach of alternatives including mobile clinics and contingency management (48).

Despite predicted potential for generalizability, this analysis is limited by both conceptual and practical challenges. Regional diversity, data availability and quality, risk of ecological fallacy, and resource variations across Central-Appalachian subregions accentuate the need for further exploration of practicality and long-term sustainability.

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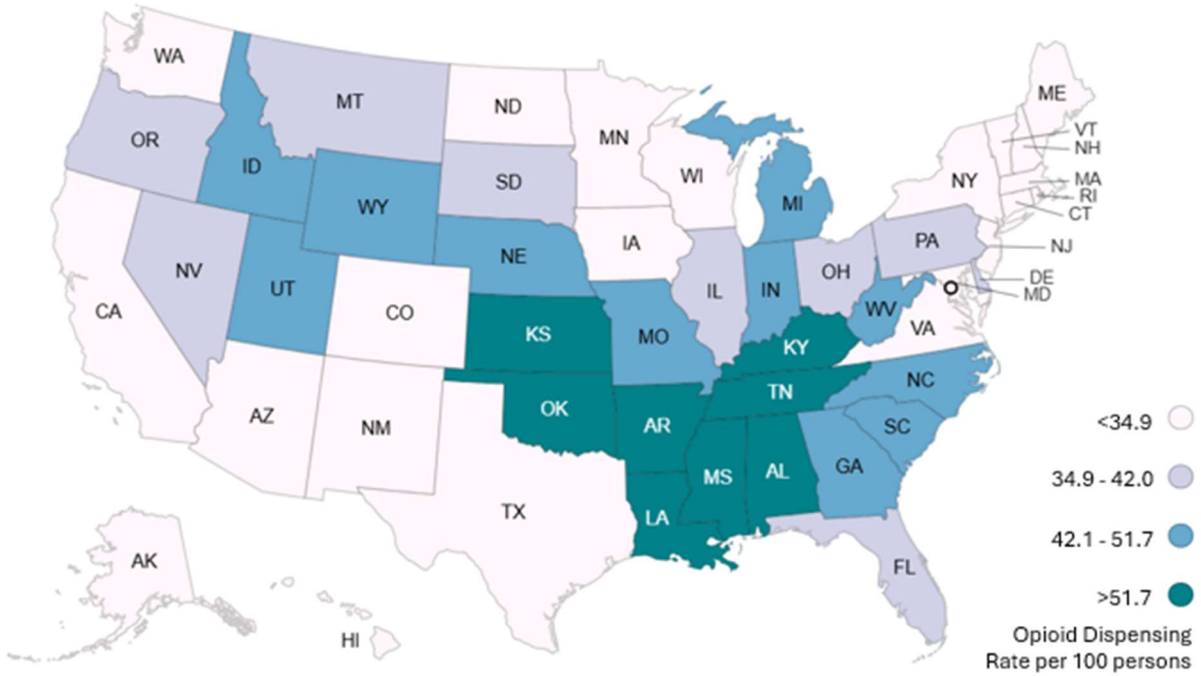
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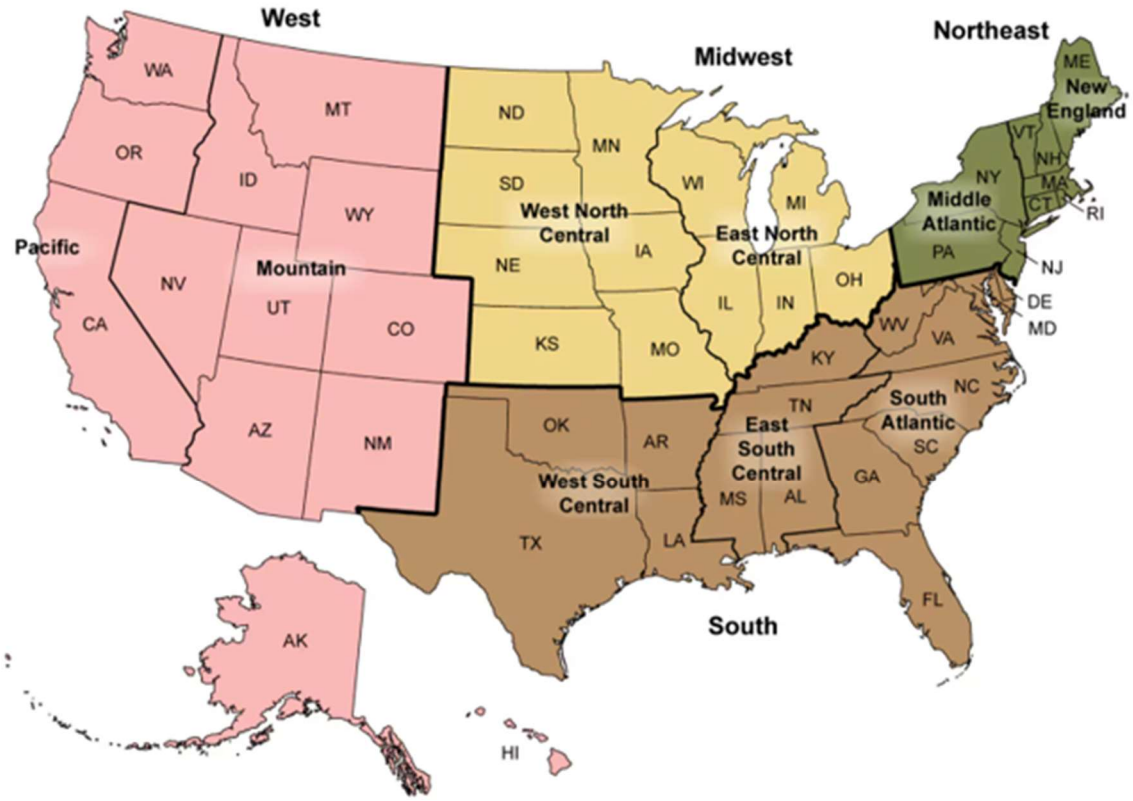
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Appendix A

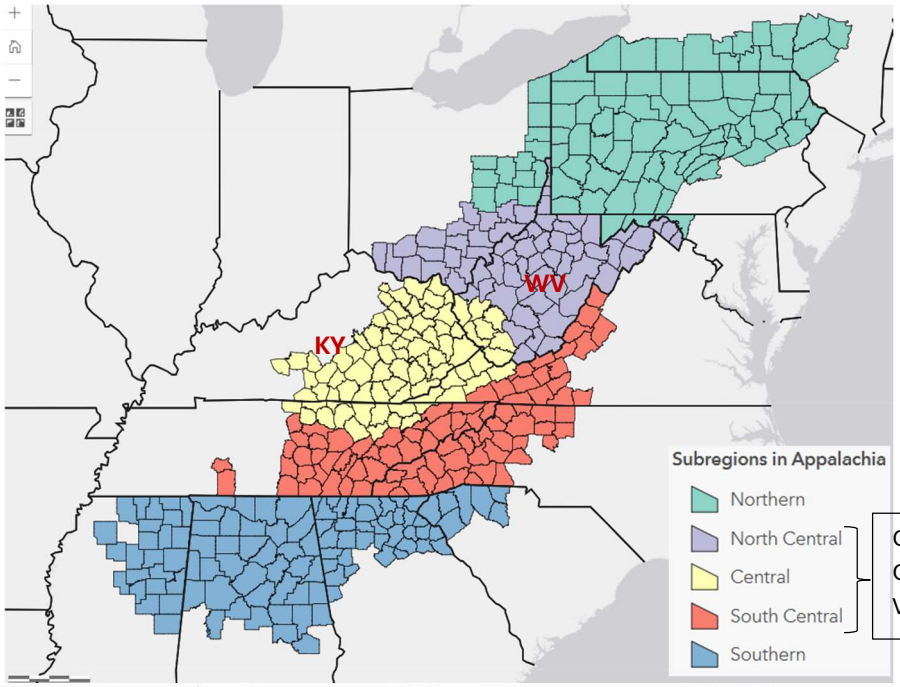
Maps Of U.S. State Opioid Dispensing Rates and Geographic Region: CDC maps showing opioid dispensing rate by Sate in 2023 and U.S. Geographical division by region(13,14).



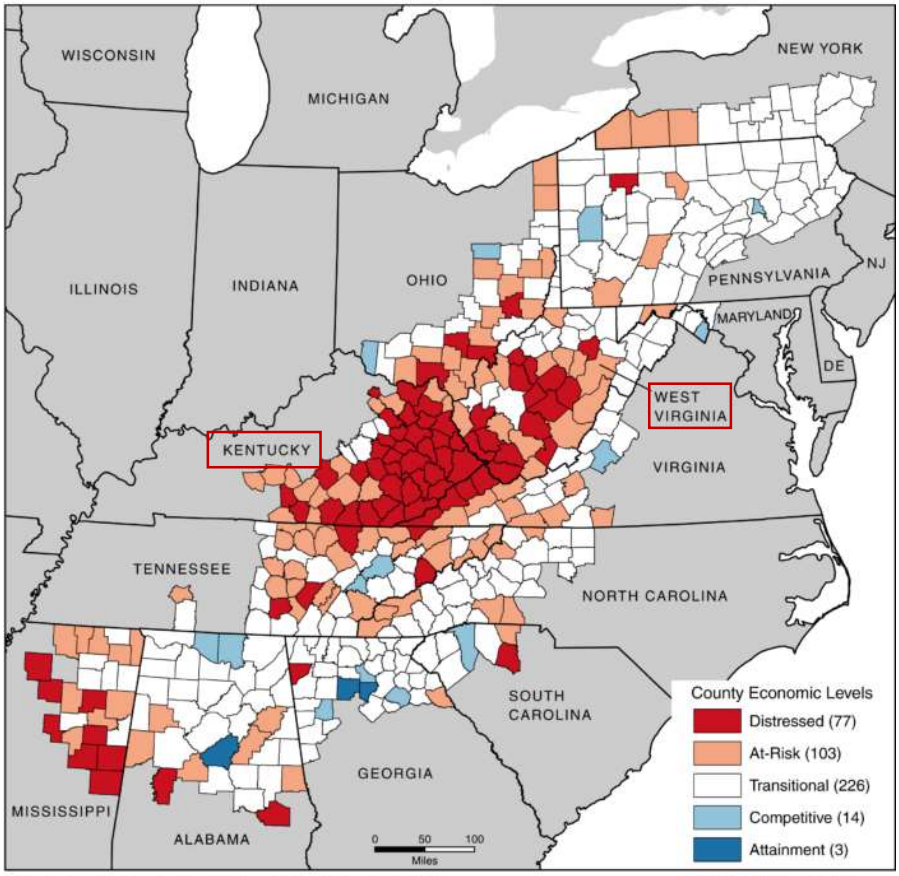


Appendix B

Maps of the U.S. Appalachian Region: Appalachian Regional Commission (ARC) maps indicating the Appalachian region and economic status and distress in fiscal year 2025 (15,16).



Georgia; Kentucky; North Carolina; Tennessee; Virginia; West-Virginia

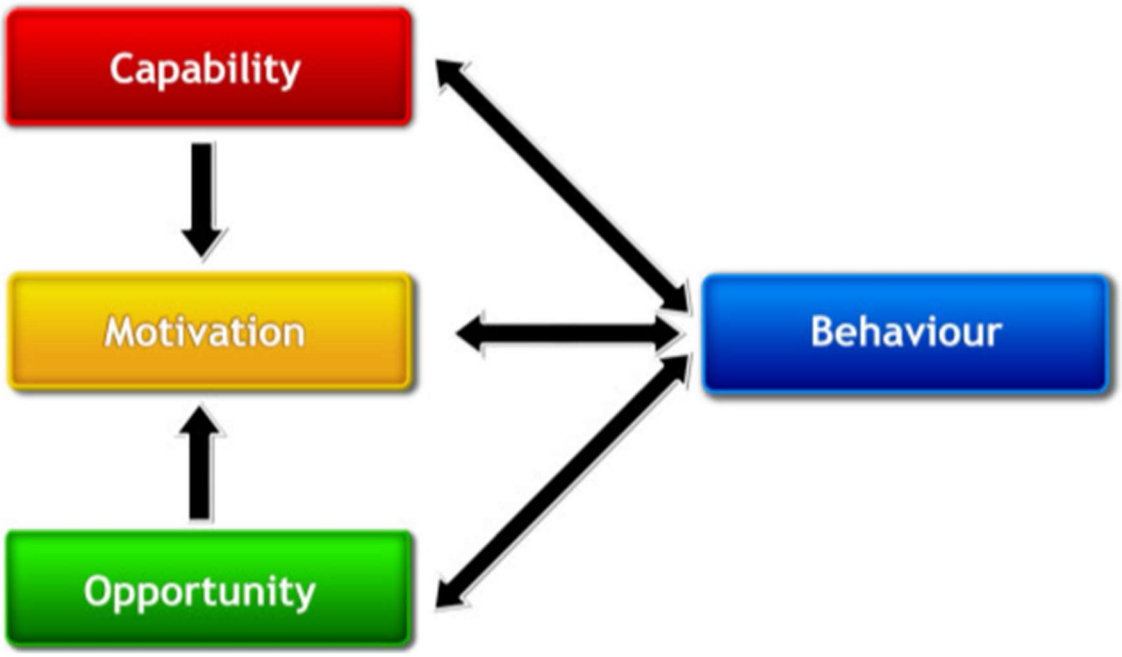


Created by the Appalachian Regional Commission, June 2024
Data Sources:
Unemployment data: U.S. Bureau of Labor Statistics, LAUS, 2020–2022
Income data: U.S. Bureau of Economic Analysis, LAPI, 2022
Poverty data: U.S. Census Bureau, American Community Survey, 2018–2022

Effective October 1, 2024 through September 30, 2025

Appendix C

The COM-B Model Framework: The COM-B system - a framework for understanding behaviour (32).



Appendix D

The TDF: The Theoretical Domains Framework with definitions and component constructs (39).

TDF Version 2	
Domain (definition)	Constructs
1. Knowledge (An awareness of the existence of something)	Knowledge (including knowledge of condition/scientific rationale) Procedural knowledge Knowledge of task environment
2. Skills (An ability or proficiency acquired through practice)	Skills Skills development Competence Ability Interpersonal skills Practice Skill assessment
3. Social/professional role and identity (A coherent set of behaviours and displayed personal qualities of an individual in a social or work setting)	Professional identity Professional role Social identity Identity Professional boundaries Professional confidence Group identity Leadership Organisational commitment
4. Beliefs about capabilities (Acceptance of the truth, reality or validity about an ability, talent or facility that a person can put to constructive use)	Self-confidence Perceived competence Self-efficacy Perceived behavioural control Beliefs Self-esteem Empowerment Professional confidence
5. Optimism (The confidence that things will happen for the best or that desired goals will be attained)	Optimism Pessimism Unrealistic optimism Identity
6. Beliefs about Consequences (Acceptance of the truth, reality, or validity about outcomes of a behaviour in a given situation)	Beliefs Outcome expectancies Characteristics of outcome expectancies Anticipated regret Consequents
7. Reinforcement (Increasing the probability of a response by arranging a dependent relationship, or contingency, between the response and a given stimulus)	Rewards (proximal/distal, valued/not valued, probable/improbable) Incentives Punishment Consequents Reinforcement Contingencies Sanctions
8. Intentions (A conscious decision to perform a behaviour or a resolve to act in a certain way)	Stability of intentions Stages of change model Transtheoretical model and stages of change
9. Goals (Mental representations of	Goals (distal/proximal) Goal priority

outcomes or end states that an individual wants to achieve)	Goal/target setting Goals (autonomous/controlled) Action planning Implementation intention
10. Memory, attention and decision processes (The ability to retain information, focus selectively on aspects of the environment and choose between two or more alternatives)	Memory Attention Attention control Decision making Cognitive overload/tiredness
11. Environmental context and resources (Any circumstance of a person's situation or environment that discourages or encourages the development of skills and abilities, independence, social competence and adaptive behaviour)	Environmental stressors Resources/material resources Organisational culture/climate Salient events/critical incidents Person × environment interaction Barriers and facilitators
12. Social influences (Those interpersonal processes that can cause individuals to change their thoughts, feelings, or behaviours)	Social pressure Social norms Group conformity Social comparisons Group norms Social support Power Intergroup conflict Alienation Group identity Modelling
13. Emotion (A complex reaction pattern, involving experiential, behavioural, and physiological elements, by which the individual attempts to deal with a personally significant matter or event)	Fear Anxiety Affect Stress Depression Positive/negative affect Burn-out
14. Behavioural regulation (Anything aimed at managing or changing objectively observed or measured actions)	Self-monitoring Breaking habit Action planning

Appendix E

The BCW: The Behavioural Change Wheel (32).

